

Unit-3

Matrix decomposition

Quadratic forms

A quadratic form of on $\mathbb{R}^n$ is a function Q on $\mathbb{R}^n$ whose value at any vector x on $\mathbb{R}^n$ can be computed by an expression $Q(x) = x^T A x$ where A is $n \times n$ symmetric matrix and it is called matrix of quadratic form.

Example:

$$\begin{aligned} \text{The simple ex for quadric form is } Q(x) &= x^T I x \\ &= x^T x \\ &= x \cdot x \\ &= \|x\|^2 \end{aligned}$$

The following examples shows the connection between any symmetric matrix A and quadratic form $x^T A x$

Problems

1. Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$ Compute $x^T A x$

$$\Rightarrow x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= 4x_1^2 + 3x_2^2$$

2) $A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ Compute $x^T A x$

$$\Rightarrow x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{aligned} &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3x_1 & -2x_2 \\ -2x_1 & 7x_2 \end{bmatrix} = \begin{bmatrix} 3x_1^2 & -2x_1x_2 & -2x_1x_2 & 7x_2^2 \end{bmatrix} \\ &= 3x_1^2 - 4x_1x_2 + 7x_2^2 \end{aligned}$$

Note:

If the quadratic form contains n variables $x_1, x_2, \dots, x_n$,
 then $A = \begin{bmatrix} \text{coeff of } x_1^2 & \frac{1}{2} \text{coeff of } x_1 x_2 & \frac{1}{2} \text{coeff of } x_1 x_3 & \dots & \frac{1}{2} \text{coeff of } x_1 x_n \\ \frac{1}{2} \text{coeff of } x_2 x_1 & \text{coeff of } x_2^2 & \frac{1}{2} \text{coeff of } x_2 x_3 & \dots & \frac{1}{2} \text{coeff of } x_2 x_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{2} \text{coeff of } x_n x_1 & \frac{1}{2} \text{coeff of } x_n x_2 & \dots & \dots & \text{coeff of } x_n^2 \end{bmatrix}$

Ex1

1. Let $Q(x) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$

→ Matrix of quadratic form $A = \begin{bmatrix} 5 & -1/2 & 0 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix}$

change of variable in quadratic form:-

If X represents a variable in R^n , then a change of variable is an equation of the form $X = PY$ or equivalently $Y = P^{-1}X$ for some invertible matrix P and for some new variable Y .

If the change of variable is made in the quadratic form then

$$X^T A X = (PY)^T A (PY)$$

$$= Y^T (P^T A P) Y \quad \text{--- (1)}$$

if P orthogonally diagonalizes A , $P^T = P^{-1}$ and then eq (1)

simplifies $X^T A X = Y^T (P^T A P) Y$

$= (Y^T D Y)$ which is called Canonical form

27-03-26

Note:

In orthogonal diagonalization P is the matrix whose columns are orthonormal normalized eigen vectors

pure

1). Reduce the quadratic form into canonical form and Mention change of variable

a) $x_1^2 - 8x_1x_2 - 5x_2^2$

$$A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$$

characteristic equation of A is $\lambda^2 - \text{tr}(A)\lambda + |A| = 0$

$$\text{i.e., } \lambda^2 + 4\lambda - 21 = 0$$

$$(\lambda + 7)(\lambda - 3) = 0$$

$$\Rightarrow \lambda_1 = -7, \lambda_2 = 3$$

$$\therefore D = \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix}$$

Let the change of variable be $X = PY$,
to find P consider $[A - \lambda I]X = 0$

$$\text{i.e. } \begin{bmatrix} 1-\lambda & -4 \\ -4 & -5-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

Put $\lambda = -7$ in eq (1)

$$\text{i.e. } \begin{bmatrix} 8 & -4 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 8x - 4y &= 0 & \text{--- (2)} \\ -4x + 2y &= 0 & \text{--- (3)} \end{aligned} \quad \left\{ \begin{array}{l} \text{same} \\ \text{consider any 1} \end{array} \right.$$

$$\text{From (3)} \Rightarrow -4x = -2y$$

$$2x = y$$

$$\Rightarrow \frac{x}{1} = \frac{y}{2}$$

$$\therefore \text{Eigen vector } X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\therefore \text{Normalized eigen vector is } \hat{X}_1 = \frac{X_1}{\|X_1\|}$$

$$\|X_1\| = \sqrt{1^2 + 2^2} = \sqrt{5} \quad \Rightarrow \left[\frac{X_1}{\sqrt{5}} \right] = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

Put $\lambda = 3$ in eq (1)

$$\text{i.e. } \begin{bmatrix} -2 & -4 \\ -4 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -2x - 4y &= 0 & \text{--- (4)} \\ -4x - 8y &= 0 & \text{--- (5)} \end{aligned} \quad \left\{ \begin{array}{l} \text{same} \end{array} \right.$$

$$\begin{aligned} \text{from (4)} \Rightarrow -2xy &= 4y \\ -2x &= 4y \\ -x &= 2y \\ \frac{x}{y} &= \frac{y}{-1} \end{aligned}$$

$$\therefore x_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\|x\| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\therefore \text{Normalized eigen vector is } \hat{x}_2 = \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix}$$

$$\therefore P = [\hat{x}_1 \hat{x}_2] = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix}$$

Since $X = PY$ we have $X^T A X = Y^T D Y$

$$= \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} -7y_1 \\ 3y_2 \end{bmatrix}$$

$$= \underline{-7y_1^2 + 3y_2^2} \text{ which is called canonical form}$$

change of variable be $X = PY$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ 2/\sqrt{5} & -1/\sqrt{5} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5}y_1 + 2/\sqrt{5}y_2 \\ 2/\sqrt{5}y_1 - 1/\sqrt{5}y_2 \end{bmatrix}$$

$$x_1 = \underline{1/\sqrt{5}y_1 + 2/\sqrt{5}y_2}$$

$$x_2 = \underline{2/\sqrt{5}y_1 - 1/\sqrt{5}y_2}$$

2) $3x_1^2 - 4x_1x_2 + 6x_2^2$ Reduce the given quadratic form into canonical form and mention change of variable

$$\rightarrow A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}$$

18-4

characteristic equation of A is $\lambda^2 - \text{tr}(A)\lambda + |A| = 0$

$$\text{i.e. } \lambda^2 - 9\lambda + 14 = 0$$

$$(\lambda - 7)(\lambda - 2) = 0$$

$$\Rightarrow \lambda_1 = 7 \quad \lambda_2 = 2$$

$$\therefore D = \begin{bmatrix} 7 & 0 \\ 0 & 2 \end{bmatrix}$$

Let the change of variable be $X = PY$,

to find P consider $(A - \lambda I)X = 0$

$$\text{i.e. } \begin{bmatrix} 3-\lambda & -2 \\ -2 & 6-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

Put $\lambda = 7$ in eq (1)

$$\text{i.e. } \begin{bmatrix} -4 & -2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-4x - 2y = 0 \quad \text{--- (2)}$$

$$-2x - y = 0 \quad \text{--- (3)}$$

} same

from (2) $-4x = 2y$

$$-2x = y$$

$$\frac{x}{1} = \frac{y}{-2}$$

$$\therefore \text{Eigenvector } X_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\therefore \text{Normalized eigenvector is } \hat{X}_1 = \begin{bmatrix} \frac{X_1}{\|X_1\|} \end{bmatrix} \quad \sqrt{1^2 + 4^2} = \sqrt{5}$$

$$= \begin{bmatrix} \frac{X_1}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}$$

Put $\lambda = 2$ in eq. ①

$$\text{i.e. } \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$1x - 2y = 0 \quad \text{--- ④}$$

$$-2x + 4y = 0 \quad \text{--- ⑤}$$

} same

from ④ $1x = 2y$

$$\frac{x}{2} = y$$

$$x_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\hat{x}_2 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$\therefore P = [\hat{x}_1 \quad \hat{x}_2] = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

Since $x = PY$ we have $x^T A x = y^T D y$

$$= \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 7y_1 \\ 2y_2 \end{bmatrix}$$

$$= 7y_1^2 + 2y_2^2 \quad \text{or canonical form}$$

change of variable let $x = PY$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5}y_1 + 2/\sqrt{5}y_2 \\ -2/\sqrt{5}y_1 + 1/\sqrt{5}y_2 \end{bmatrix}$$

$$x_1 = 1/\sqrt{5}y_1 + 2/\sqrt{5}y_2 \quad x_2 = -2/\sqrt{5}y_1 + 1/\sqrt{5}y_2$$

c) $x_1^2 - 6x_1x_2 + 9x_2^2$

$$A = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix}$$

Characteristic equation of A is $\lambda^2 - \text{tr}(A)\lambda + |A| = 0$

$$\lambda^2 - 10\lambda + 0 = 0$$

$$\lambda^2 = 10\lambda$$

$$\lambda = 10, 0$$

$$\therefore D = \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$$

$$X = PY$$

To find P:

Consider $(A - \lambda I)X = 0$

$$\begin{bmatrix} 1-\lambda & -3 \\ -3 & 9-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

~~9x~~ Put $\lambda = 10$ in (1)

$$\begin{bmatrix} -9 & -3 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} -9x - 3y = 0 \\ -3x - y = 0 \end{cases} \text{ same}$$

$$-3x = y$$

$$\frac{x}{1} = \frac{y}{-3}$$

Eigen vector $X_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ $\|X_1\| = \sqrt{1+9} = \sqrt{10}$

Normalized eigen vector is $\hat{X}_1 = \begin{bmatrix} 1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix}$

Put $\lambda=0$ in ①

$$\begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} x-3y &= 0 \\ -3x+9y &= 0 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{same}$$

$$x=3y$$

$$\frac{x}{3} = \frac{y}{1}$$

$$\text{EV in } x_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \|x_2\| = \sqrt{9+1} = \sqrt{10}$$

$$\text{Normalized eigen vector, } \hat{x}_2 = \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$$

$$P = [\hat{x}_1 \quad \hat{x}_2] = \begin{bmatrix} 1/\sqrt{10} & 3/\sqrt{10} \\ -3/\sqrt{10} & 1/\sqrt{10} \end{bmatrix}$$

Since $x=Py$ we have $x^T A x = y^T D y$

$$= [y_1 \quad y_2] \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= [y_1 \quad y_2] \begin{bmatrix} 10y_1 \\ 0y_2 \end{bmatrix}$$

$$= 10y_1^2 + 0y_2^2$$

$= 10y_1^2$ is the canonical form.

change of variable be $x=Py$

$$\text{i.e. } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{10} & 3/\sqrt{10} \\ -3/\sqrt{10} & 1/\sqrt{10} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$x_1 = \frac{1}{\sqrt{10}} (y_1 + 3y_2)$$

$$x_2 = \frac{1}{\sqrt{10}} (-3y_1 + y_2)$$

Classification of Quadratic forms

A quadratic form $Q(x)$ is

- 1) Positive definite if $Q(x) > 0 \quad \forall x \neq 0$
- 2) Negative definite if $Q(x) < 0, \forall x \neq 0$
- 3) Indefinite if $Q(x)$ assumes both positive and negative values

Quadratic forms and eigen values

Theorem: Let A be $n \times n$ symmetric matrix, then the quadratic form, $x^T A x$ is

- 1) Positive definite iff all the eigen values of A are positive
- 2) Negative definite iff the eigen values of A are all negative
- 3) Indefinite iff A has both positive and negative eigen values.

Problems

- 1) check whether the following quadratic forms are positive definite, negative definite, indefinite

i] $Q(x) = 3x_1^2 + 0x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3$

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

characteristic eqn of $A = \lambda^3 - \text{tr}(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

$$\text{tr}(A) = 3 + 0 + 1 = 4$$

$$A_{11} + A_{22} + A_{33} = \begin{vmatrix} 2 & 2 \\ 2 & 1 \end{vmatrix} + \begin{vmatrix} 3 & 0 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 3 & 2 \\ 2 & 2 \end{vmatrix}$$

$$(2-4) + (3-0) + (6-4)$$

$$-2 + 3 + 2$$

$$3$$

$$|A| = \begin{vmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{vmatrix}$$

$$= 3(2-4) - 2(2-0) + 0(4-0)$$

$$= 3(-2) - 4 + 0$$

$$= -6 - 4$$

$$= -10$$

$$\lambda^3 - 6\lambda^2 + 3\lambda + 10 = 0$$

$$\lambda_1 = -1, 5, 2$$

are

Eigen values are both positive & negative, therefore given quadratic form is indefinite.

2). $Q(x) = 3x_1^2 + 5x_2^2 + 3x_3^2 - 2x_1x_2 + 0x_1x_3 - 2x_2x_3$

$$A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

characteristic eqn is $\lambda^3 - k(A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

$$k(A) = 11$$

$$A_{11} + A_{22} + A_{33} = \begin{vmatrix} 5 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & -1 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & -1 \\ -1 & 5 \end{vmatrix}$$

$$= (15-1) + (9-1) + (15-1)$$

$$= 14 + 8 + 14$$

$$= 36$$

$$|A| = \begin{vmatrix} 3 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 3 \end{vmatrix}$$

$$= 3(15-1) + 1(-3+1) + 1(1-5)$$

$$= 3(14) + 1(-2) + 1(-4)$$

$$= 42 - 2 - 4$$

$$= 36$$

$$\lambda^3 - 26\lambda^2 + 36\lambda - 36 = 0$$

$$\lambda = 6, 3, 2$$

$\therefore$ Eigen values are both positive & negative.

Since both eigen values are positive quadratic form is positive definite.

iii) $-x_1^2 - 2x_2^2 - 3x_3^2 + 2x_1x_2$

iv) $2x_1x_2 + 2x_1x_3 - 2x_2x_3$

QR factorization

* Theorem: If A is $m \times n$ matrix with linearly independent columns then A can be factored as $A = QR$, where Q is $m \times n$ matrix whose columns form an orthonormal basis for column space of A and R is $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

Problems

1) Obtain QR factorization of matrix $A = \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$

$\rightarrow$ We know that columns of A form a basis $\{x_1, x_2\}$ for column space of A , where $x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $x_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

first we find orthogonal basis for column space of A using gram schmidt's process.

$$v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$v_2 = x_2 - \left(\frac{x_2 \cdot v_1}{v_1 \cdot v_1} \right) \cdot v_1$$

$$v_2 \cdot v_1 = (-1 \times 1 + 0) = (-1)$$

$$v_1 \cdot v_1 = (1 \times 1) + (-1 \times -1) = 2$$

$$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \frac{(-1)}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 \\ -1/2 \end{bmatrix}$$

$\therefore \{v_1, v_2\}$ forms orthogonal basis for $\text{col}(A)$.

Hence $\left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|} \right\}$ forms orthonormal basis for column space of A .

$$\|v_1\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$\|v_2\| = \sqrt{(-1/2)^2 + (-1/2)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{2}{4}} = \frac{\sqrt{2}}{2}$$

$$\therefore Q = \begin{bmatrix} \frac{v_1}{\|v_1\|} & \frac{v_2}{\|v_2\|} \end{bmatrix}$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$\frac{-1/2}{1/\sqrt{2}} = \frac{-1/2 \times \sqrt{2}}{1} = \frac{-\sqrt{2}}{2} = \frac{-1}{\sqrt{2}}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{bmatrix}$$

=

$$\text{Then } \underline{A = QR}$$

Q. Obtain QR factorization of matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$$\rightarrow \text{Let } x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

W.K.T $\{x_1, x_2, x_3\}$ forms a basis for $\text{col}(A)$. Therefore to get orthogonal basis $\{v_1, v_2, v_3\}$, we use gram schmidt's process

$$v_1 = x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$v_2 = x_2 - \left(\frac{x_2 \cdot v_1}{v_1 \cdot v_1} \right) v_1$$

$$x_2 \cdot v_1 = (1) + (0) + (0) = 1$$

$$v_1 \cdot v_1 = (1) + (1) + (0) = 2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

$$x_3 \cdot v_1 = 0 + 1 + 0 = 1$$

$$= \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

$$v_3 = x_3 - \left(\frac{x_3 \cdot v_1}{v_1 \cdot v_1} \right) v_1 - \left(\frac{x_3 \cdot v_2}{v_2 \cdot v_2} \right) v_2$$

$$x_3 \cdot v_2 = (0) + (1 \times \frac{1}{2}) + 1$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{\frac{1}{2}}{\frac{3}{2}} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

$$-\frac{1}{2} \times \frac{1}{2}$$

$$\frac{1}{2} \times \frac{1}{2}$$

$$\frac{1}{2} \times \frac{1}{2}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{6} \\ \frac{1}{3} \end{bmatrix}$$

$$0 - \frac{1}{2} - \frac{1}{6} = -\frac{3}{6} - \frac{1}{6} = -\frac{4}{6}$$

$$1 - \frac{1}{2} + \frac{1}{6} = \frac{6}{6} - \frac{3}{6} + \frac{1}{6} = \frac{4}{6}$$

$$1 - \frac{1}{2} + \frac{1}{6}$$

$$\frac{1}{2} + \frac{1}{6} = \frac{2}{3}$$

$$\underline{\underline{\begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix}}}$$

$\therefore \{v_1, v_2, v_3\}$ forms orthogonal basis for $\text{col}(A)$
Hence $\left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \frac{v_3}{\|v_3\|} \right\}$ forms orthonormal basis for $\text{col}(A)$

$$\|v_1\| = \sqrt{1^2 + 1^2 + 0} = \underline{\underline{\sqrt{2}}}$$

$$\|v_2\| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + (-1)^2} = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{6}{4}} = \underline{\underline{\frac{\sqrt{3}}{2}}}$$

$$\|v_3\| = \sqrt{\left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{12}{9}} = \sqrt{\frac{4}{3}} = \underline{\underline{\frac{2}{\sqrt{3}}}}$$

$$Q = \begin{bmatrix} \frac{v_1}{\|v_1\|} & \frac{v_2}{\|v_2\|} & \frac{v_3}{\|v_3\|} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$R = Q^T A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}$$

3) Find QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

*Row
complete*

Let $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$

∴ $\{x_1, x_2, x_3\}$ forms a basis for $\text{col}(A)$ ∴ to get orthogonal vectors basis $\{v_1, v_2, v_3\}$ we use gram schmidt's process

$$V_1 = x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$x = 0 + (1)u + (1)v + 3$$

$$V_2 = x_2 - \left(\frac{x_2 \cdot V_1}{V_1 \cdot V_1} \right) V_1$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 3/4 \\ 3/4 \\ 3/4 \\ 3/4 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$1 - \frac{3}{4} = \frac{1}{4}$$

$$\|V_2\| = \frac{\sqrt{3}}{2}$$

$$V_3 = x_3 - \left(\frac{x_3 \cdot V_1}{V_1 \cdot V_1} \right) V_1 - \left(\frac{x_3 \cdot V_2}{V_2 \cdot V_2} \right) V_2$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{12}{16} \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$\frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$$

$$\frac{12}{16}$$

$$\frac{0}{16} + \frac{1}{16} - \frac{1}{16} + \frac{1}{16}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{3} \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

$$\frac{1}{2} - \frac{1}{2} = 0$$

$$\|V_3\| = \frac{\sqrt{6}}{3}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2/4 \\ 2/4 \\ 2/4 \\ 2/4 \end{bmatrix} - \begin{bmatrix} 1/2 \\ +1/6 \\ +1/6 \\ +1/6 \end{bmatrix} = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix}$$

$$-\frac{2}{3} + \frac{1}{3} = -\frac{1}{3}$$

$$-\frac{2}{3} + \frac{1}{3} = -\frac{1}{3}$$

$$Q = \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ 1/2 & 1/\sqrt{3} & 2/\sqrt{6} \\ 1/2 & 1/\sqrt{3} \times 2 & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{3} \times 2 & 1/\sqrt{6} \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} 1/2 & 1/2 & 1/2 \\ -\sqrt{3}/2 & 1/\sqrt{3} & 1/\sqrt{3} \\ 0 & 1/\sqrt{6} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3/2 & 1 \\ 0 & \sqrt{3}/2 & 1/\sqrt{3} \\ 0 & 0 & 2/\sqrt{6} \end{bmatrix}$$

VIMP

Least square problems

Least squares method is used to find the solution of inconsistent system $AX=B$.

When a solution does not exist, the best one can do is to find the value of X that makes AX as close as possible to B .

The general least square problem is to find an X that makes $\|B-AX\|$ as small as possible.

Solution of general least square problems:-

The least square solution of the equation $AX=B$, is the non-empty set of solutions which satisfies the normal equations

$$[A^T A X = A^T B] \text{ and } \hat{X} = (A^T A)^{-1} A^T B$$

1. Find the least square solution of the inconsistent system

$$AX=B, A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

→ Least square solution satisfies the normal equation $A^T A X = A^T B$

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$\therefore A^T A X = A^T B$$

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$17x + y = 19$$

$$x + 5y = 11$$

$$x=1, y=2 \quad \text{L.S solution is } \hat{X} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

2 Find the least square solution of $AX=B$, $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$

→ $A^T A x = A^T B$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 3 \\ 5 & 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 10 \\ 8 & 20 & 26 \\ 10 & 26 & 38 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 1 & 1 & 3 \\ 5 & 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix} = \begin{bmatrix} 12 \\ 12 \\ 20 \end{bmatrix}$$

$A^T A x = A^T B$

$$\begin{bmatrix} 4 & 8 & 10 \\ 8 & 20 & 26 \\ 10 & 26 & 38 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 12 \\ 12 \\ 20 \end{bmatrix}$$

① $4x + 8y + 10z = 12$

② $8x + 20y + 26z = 12$

③ $10x + 26y + 38z = 20$

$x = 10$ $y = -6$ $z = 2$

For exam do gauss elimination method

L.S solution is $\hat{x} = \begin{bmatrix} 10 \\ -6 \\ 2 \end{bmatrix}$

$$\begin{bmatrix} 4 & 8 & 10 & : & 12 \\ 8 & 20 & 26 & : & 12 \\ 10 & 26 & 38 & : & 20 \end{bmatrix}$$

$$\sim \begin{bmatrix} 4 & 8 & 10 & : & 12 \\ 0 & 4 & 6 & : & -12 \\ 0 & 12 & 26 & : & -20 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow 2R_3 - 5R_1 \end{array}$$

$$\sim \begin{bmatrix} 4 & 8 & 10 & : & 12 \\ 0 & 4 & 6 & : & -12 \\ 0 & 0 & 8 & : & 16 \end{bmatrix} \quad R_3 \rightarrow R_3 - 3R_2$$

$$= [0 : C]$$

$$4x + 8y + 10z = 12 \quad \text{--- (1)}$$

$$4y + 6z = -12 \quad \text{--- (2)}$$

$$8z = 16 \quad \text{--- (3)}$$

$$4y + 6z = -12$$

$$4y = -12 - 6z$$

$$4y = -48$$

$$y = -12$$

$$(3) \Rightarrow \boxed{z = 2}$$

$$(2) \Rightarrow \boxed{y = -6}$$

$$(1) \Rightarrow \boxed{x = 10}$$

$$4y + 12 = -12$$

$$4y = -12 - 12$$

$$4y = -24$$

$$y = -6$$

$$\therefore \text{L.S solution is } x = \begin{bmatrix} 10 \\ -6 \\ 2 \end{bmatrix}$$

3 Find the least square solution of $AX=B$, $A = \begin{bmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 5 \\ -3 \\ -5 \end{bmatrix}$

$$\rightarrow A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -3 & 5 & 7 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -9 & 0 \\ 9 & 83 & 28 \\ 0 & 28 & 14 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 1 & 1 & 1 \\ -3 & 5 & 7 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \\ -5 \end{bmatrix} = \begin{bmatrix} -3 \\ -65 \\ -28 \end{bmatrix}$$

$$\therefore A^T A x = A^T B$$

$$= \begin{bmatrix} 3 & -9 & 0 \\ 9 & 83 & 28 \\ 0 & 28 & 14 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ -65 \\ -28 \end{bmatrix}$$

$$= 3x - 9y = -3$$

$$9x + 83y + 28z = -65$$

$$28y + 14z = -28$$

$$\begin{bmatrix} 3 & 9 & 0 & : & -3 \\ 9 & 83 & 28 & : & -65 \\ 0 & 28 & 14 & : & -28 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & 9 & 0 & : & -3 \\ 0 & 56 & 28 & : & -56 \\ 0 & 28 & 14 & : & -28 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \end{array}$$

$$\sim \begin{bmatrix} 3 & 9 & 0 & : & -3 \\ 0 & 56 & 28 & : & -56 \\ 0 & 0 & 0 & : & 0 \end{bmatrix} \begin{array}{l} \\ R_3 \rightarrow 2R_3 - R_2 \end{array}$$

x and y are the basic variables since it is pivot column
 z is the free variable.

$$3x + 9y + 0z = -3 \Rightarrow 3x = -3 - 9y$$

Put $y = -1 - 3/2 z$ in ①

$$x = -1 - 3y/2 \quad \text{--- ①}$$

$$56y + 28z = -56 \Rightarrow y = \frac{-56 - 28z}{56}$$

$$x = -1 - 3(-1 - 3/2 z)$$

$$y = -1 - 3/2 z$$

$$x = 2 + 3z/2$$

z is free

$$\therefore \text{General solution } \hat{x} = \begin{bmatrix} 2 + 3z/2 \\ -1 - 3/2 z \\ z \end{bmatrix}$$

To get specific least square solution Put $z=0$

$$\text{L.S solution } \hat{x} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

4. Find the least square solution of $AX=B$, $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix}$

→ Least square solution is given by solving the normal eq

$$A^T A X = A^T B \quad \text{--- (1)}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 10 \\ 1 & 10 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix} = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}$$

$$A^T A X = A^T B$$

$$\begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}$$

$$4x + 2y + 2z = 14$$

$$2x + 2y + 0z = 4$$

$$2x + 0y + 2z = 10$$

Gauss elimination

$$\sim \begin{bmatrix} 4 & 2 & 2 & : & 14 \\ 2 & 2 & 0 & : & 4 \\ 2 & 0 & 2 & : & 10 \end{bmatrix}$$

$$\sim \begin{bmatrix} 4 & 2 & 2 & : & 14 \\ 0 & 2 & -2 & : & -6 \\ 0 & -2 & 2 & : & 6 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow 2R_2 - R_1 \\ R_3 \rightarrow 2R_3 - R_1 \end{array}$$

$$\begin{bmatrix} 4 & 2 & 2 & : & 14 \\ 0 & 2 & -2 & : & -6 \\ 0 & 0 & 0 & : & 0 \end{bmatrix} \quad \begin{array}{l} -? \\ R_3 \rightarrow R_3 + R_2 \end{array}$$

$\sim [0:0]$

$\text{Rank}(A) = 2$ $\text{Rank}(A:B) = 2$ no of unknown = 3

x and y are basic elements, z is a free variable.

$z = 3$

$$4x + 2y + 2z = 14 \Rightarrow x = \frac{14 - 2y - 2z}{4} \quad \text{--- (1)}$$

$$z = \frac{14 - 2(-3+3) - 23}{4}$$

$$2y - 2z = -6 \Rightarrow y = \frac{-6 + 2z}{2} \Rightarrow -3 + z \quad \text{--- (2)}$$

$$\frac{14 + 6 - 2 - 23}{4}$$

$$\frac{14 + 22 - 23}{4}$$

$$\frac{7 - 2 - 3}{4}$$

$$\frac{2}{4}$$

$$\frac{7 - (-3+3) - 3}{4}$$

$$\frac{7 + 3 - 23}{4}$$

$$\frac{10 - 23}{4}$$

$$\frac{3}{4}$$

$$\frac{5 - 3}{4}$$

$$\frac{2}{4}$$

$$C.S = \begin{bmatrix} 15 - z \\ -3 + z \\ z \end{bmatrix}$$

To get specific least square solution Put $z=0$

$$L.S \text{ solution } \hat{x} = \begin{bmatrix} 5 \\ -3 \\ 0 \end{bmatrix}$$

5 Find the least square solution of $AX=B$ for $A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

> Least square solution is given by solving the normal eq $A^T A x = A^T B$ --- (1)

$$A^T A = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}$$

$$A^T B = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$A^T A x = A^T B$$

$$\begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$6x - 11y = -4$$

$$-11x + 22y = 11$$

$$x = 3 \quad y = 2$$

$$\hat{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

QR factorization $\rightarrow$ 1 ques sure
 SVD $\rightarrow$ 1 sure
 least square $\rightarrow$ 1 sure

Unit III
 study these 3 problems

M	T	W	T	F	S	S
Page No.						YOUVA
Date						

V_{imp}
 1 sure

SVD Singular Value Decomposition

Let A be $m \times n$ matrix, the singular values of A are the square roots of eigen values of $A^T A$, denoted by $\sigma_1, \sigma_2, \dots, \sigma_n$ and they are arranged in descending order i.e. $\sigma_i = \sqrt{\lambda_i}, 1 \leq i \leq n$

Theorem

Singular value decomposition theorem

Let A be an $m \times n$ matrix with Rank n then there exist an $m \times n$ matrix $\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}$ for which the diagonal entries in

D are the first n -singular values of A , $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$ and there exist an $m \times m$ orthogonal matrix U and $n \times n$ orthogonal matrix V such that $A = U \Sigma V^T$

Note: Any factorization $A = U \Sigma V^T$, with U and V orthogonal, $\Sigma = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$ and positive diagonal entries in D is called singular value decomposition of A .

2. In $A = U \Sigma V^T$, $V = [\hat{v}_1 \ \hat{v}_2 \ \hat{v}_3 \ \dots]$ where $\hat{v}_i = \frac{v_i}{\|v_i\|}$ is unit

eigen vector of $A^T A$.

$$\Sigma = D = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ 0 & 0 & \sigma_3 & \dots & 0 \\ \vdots & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \sigma_n \end{bmatrix} \text{ where } \sigma_i = \sqrt{\lambda_i}$$

$$U = [\hat{u}_1 \ \hat{u}_2 \ \dots] \text{ where } \hat{u}_i = \frac{1}{\sigma_i} A \hat{v}_i$$

Problems

6^m

1) Find the SVD of $A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$

→ SVD of $A = U \Sigma V^T$

$$A^T A = \begin{bmatrix} 2 & 0 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix}$$

characteristic eq of $A^T A$ is $\lambda^2 - \text{tr}(A^T A)\lambda + |A^T A| = 0$

$$\lambda^2 - 17\lambda + 16 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 16) = 0$$

$$\lambda = 1, 16$$

$$\lambda_1 = 16, \lambda_2 = 1 \quad [\text{descending order}]$$

$$\sigma_1 = \sqrt{16} = 4$$

$$\sigma_2 = \sqrt{1} = 1$$

$$D = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} = \Sigma$$

To find eigen value vectors of $A^T A$

consider $[A^T A - \lambda I] X = 0$

$$\begin{bmatrix} 4-\lambda & 6 \\ 6 & 13-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

Put $\lambda = 16$ in (1)

$$\begin{bmatrix} -12 & 6 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-12x + 6y = 0 \quad \text{--- (2)}$$

$$6x - 3y = 0 \quad \text{--- (3)}$$

} same

$$\textcircled{3} \Rightarrow 6x = 3y$$

$$2x = y \Rightarrow \frac{x}{1} = \frac{y}{2}$$

$$\therefore \text{Eigenvector } v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \|v_1\| = \sqrt{5}$$

$$\therefore v_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\hat{v}_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} 4/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

Put $\lambda = 1$ in ① $\begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\Rightarrow 3x + 6y = 0$$

$$x = -2y \Rightarrow \frac{x}{-2} = y$$

$$v_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \|v_2\| = \sqrt{5}$$

$$\hat{v}_2 = \frac{v_2}{\|v_2\|} = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$V = [\hat{v}_1 \quad \hat{v}_2]$$

$$V = \begin{bmatrix} 4/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

$$U = [\hat{u}_1 \quad \hat{u}_2]$$

$$\hat{u}_1 = \frac{1}{\sigma_1} A \hat{v}_1$$

$$= \frac{1}{4} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 4/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$\hat{u}_2 = \frac{1}{\sigma_2} A \hat{v}_2 = \frac{1}{1} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} = \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$U = [\hat{u}_1 \quad \hat{u}_2]$$

$$U = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$\therefore \text{SVD of } A = U \Sigma V^T$$

$$= \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$$

2) Find the SVD of $A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$

$\Rightarrow$ SVD of $A = U \Sigma V^T$

$$A^T A = \begin{bmatrix} 2 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}$$

To find ch. eq. of $A^T A$ $\lambda^2 - \text{tr}(A^T A)\lambda + |A^T A| = 0$

$$\lambda^2 - 13\lambda + 36 = 0$$

$$(\lambda - 9)(\lambda - 4) = 0$$

$$\Rightarrow \lambda = 9, 4$$

$$\sigma_1 = \sqrt{9} = 3, \sigma_2 = \sqrt{4} = 2$$

$$D = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} = \Sigma$$

To find eigen vector of $A^T A$:

Consider $[A^T A - \lambda I]x = 0$

$$\begin{bmatrix} 8-\lambda & 2 \\ 2 & 5-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

Put $\lambda = 9$ in (1)

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -x + 2y &= 0 \\ 2x - 4y &= 0 \end{aligned} \quad \text{same}$$

$$\Rightarrow \frac{x}{2} = \frac{y}{1} \quad : v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \|v_1\| = \sqrt{5}, \hat{v}_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

Put $\lambda = 4$ in (1)

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 4x + 2y &= 0 \\ 2x + y &= 0 \end{aligned} \quad \text{same}$$

$$\frac{x}{-1} = \frac{y}{2} \quad : v_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \|v_2\| = \sqrt{5}, \hat{v}_2 = \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$V = [\hat{v}_1 \hat{v}_2] = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$U = [\hat{u}_1 \hat{u}_2]$$

$$\hat{u}_1 = \frac{1}{\|Av_1\|} Av_1 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

Task-3

1) Find QR factorization of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

2) Find QR factorization of the matrix $A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 1 & -4 \\ -1 & 4 & -3 \\ 1 & -4 & 7 \\ 1 & 2 & 1 \end{bmatrix}$

3) Find the least square solution of $AX=B$ for $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$

4) Find the least square solution of $AX=B$ for $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}$

5) Check whether the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x^2, x+y, z)$ is a lin. or not.

6) Find the matrix of linear transformation defined by $T(x, y, z) = (x+y, 2z-x)$ w.r.to the basis $\beta_1 = \{(1, 0, -1), (1, 1, 1), (1, 0, 0)\}$ $\beta_2 = \{(1, 1), (1, -1)\}$

Continuation

$$u_2 = \frac{1}{\sqrt{2}} A \hat{v}_2 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$U = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

$$\therefore \text{SVD of } A = U \Sigma V^T = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$$

3) Find the SVD of $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$

$\rightarrow = U \Sigma V^T$

$$A^T A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 2 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix}$$

To find Eigen values of $A^T A$:-

ch eqn $\lambda^2 - \text{tr}(A^T A)\lambda + |A^T A| = 0$

$$\lambda^2 - 18\lambda + 0 = 0$$

$$\lambda(\lambda - 18) = 0$$

$$\lambda = 0, \lambda = 18$$

$$\Rightarrow \lambda_1 = 18, \lambda_2 = 0$$

$$\sigma_1 = \sqrt{18} = 3\sqrt{2}$$

$$\Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}_{3 \times 2}$$

1b find E.V of $A^T A$:-

$$\begin{bmatrix} 9-\lambda & -9 \\ -9 & 9-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

$$-9x - 9y = 0 \Rightarrow -9x = 9y \Rightarrow -x = y \Rightarrow \frac{x}{1} = \frac{y}{-1}$$

$$\therefore v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \|v_1\| = \sqrt{2} \Rightarrow \hat{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

Put $\lambda = 0$ in (1)

$$\begin{bmatrix} 9 & -9 \\ -9 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$9x - 9y = 0$$

$$x = y$$

$$\frac{x}{1} = \frac{y}{1} \therefore v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \|v_2\| = \sqrt{2} \Rightarrow \hat{v}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

So find $U = [\hat{u}_1 \ \hat{u}_2 \ \hat{u}_3]$

$$\hat{u}_1 = \frac{1}{\sigma_1} \hat{v}_1$$

$$= \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\hat{u}_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

Since $r_2 = 0$, the other columns of U are found by extending $\{\hat{u}_1\}$ to an orthonormal basis for $\mathbb{R}^3$, in this case we need two orthogonal unit vectors $\hat{u}_2$ and $\hat{u}_3$ that are orthogonal to $\hat{u}_1$.

Each vector must satisfy the condition $\hat{u}_1^T x = 0$

$$\Rightarrow \hat{u}_1^T x = 0$$

$$\begin{bmatrix} \frac{1}{3} & \frac{-2}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \frac{1}{3}x - \frac{2}{3}y + \frac{2}{3}z = 0$$

$$x - 2y + 2z = 0$$

Here x is a basic variable, y & z are free variable

$$\Rightarrow x = 2y - 2z$$

$$y = y \text{ (free)}$$

$$z = z \text{ (free)}$$

$$x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2y - 2z \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore$ Basis for solution set for the above equation is $w_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, w_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$

Apply gram Schmidt's process to $\{w_2, w_3\}$ to get orthogonal basis $\{u_2, u_3\}$.

$$u_2 = w_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$u_3 = w_3 - \frac{w_3 \cdot u_2}{u_2 \cdot u_2} u_2$$

$$u_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} - \frac{(-4)}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 8/5 \\ 4/5 \\ 0 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 4/5 \\ 1 \end{bmatrix}$$

$$\|u_3\| = \sqrt{4/25 + 16/25 + 1} = \sqrt{45}$$

orthogonal vector $\hat{u}_3 = \frac{u_3}{\|u_3\|} = \begin{bmatrix} -2/\sqrt{45} \\ 4/\sqrt{45} \\ 1/\sqrt{45} \end{bmatrix}$

$$\hat{u}_3 = \frac{u_3}{\|u_3\|} = \begin{bmatrix} -2/\sqrt{45} \\ 4/\sqrt{45} \\ 1/\sqrt{45} \end{bmatrix}$$

$$U = [\hat{u}_1 \hat{u}_2 \hat{u}_3]$$

$$U = \begin{bmatrix} 1/3 & 2/\sqrt{5} & -2/\sqrt{45} \\ -2/3 & 1/\sqrt{5} & 4/\sqrt{45} \\ 2/3 & 0 & 1/\sqrt{45} \end{bmatrix}$$

$$\therefore \text{SVD of } A = U \Sigma V^T = \begin{bmatrix} 1/3 & 2/\sqrt{5} & -2/\sqrt{45} \\ -2/3 & 1/\sqrt{5} & 4/\sqrt{45} \\ 2/3 & 0 & 1/\sqrt{45} \end{bmatrix} \begin{bmatrix} 3/\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ \sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

4) Find SVD of $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}_{2 \times 3}$

→ SVD of $A = U \Sigma V^T$

$$A^T A = \begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} = \begin{bmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{bmatrix}$$

$$\text{Ch. eqn} = \lambda^3 - 450\lambda^2 + (14400 + 14400 + 3600)\lambda + 80(14400) - 100(14400) + 40(7200)$$

$$= \lambda^3 - 450\lambda^2 + 32400\lambda + 0$$

$$= \lambda(\lambda^2 - 450\lambda + 32400) = 0$$

$$\lambda_1 = 360, 90, 0 \quad \sigma_1 = 6\sqrt{10} \quad \sigma_2 = 3\sqrt{10}$$

$$D = \begin{bmatrix} 6\sqrt{10} & 0 \\ 0 & 3\sqrt{10} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{bmatrix}_{2 \times 3}$$

Consider $[A^T A - \lambda I] x = 0$

$$\begin{bmatrix} 80 - \lambda & 100 & 40 \\ 100 & 170 - \lambda & 140 \\ 40 & 140 & 200 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

Put $\lambda = 360$ in (1)

$$-280x + 100y + 40z = 0 \quad \text{--- (1)}$$

$$100x - 190y + 140z = 0 \quad \text{--- (2)}$$

$$40x + 140y - 160z = 0$$

$$[A:B] = \begin{bmatrix} -280 & 100 & 40 & : & 0 \\ 100 & -190 & 140 & : & 0 \\ 40 & 140 & -160 & : & 0 \end{bmatrix}$$

Consider (1) & (2)

$$\frac{x}{\begin{vmatrix} 100 & 40 \\ -190 & 140 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} -280 & 40 \\ 100 & 140 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -280 & 100 \\ 100 & -190 \end{vmatrix}}$$

$$= \frac{x}{21600} = \frac{-y}{43200} = \frac{z}{43200}$$

$$= \frac{x}{42} = \frac{y}{1} = \frac{z}{1}$$

$$= \frac{x}{1} = \frac{y}{2} = \frac{z}{2}$$

$$\hat{v}_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = v_1$$

$$\|v_1\| = 3$$

$$\hat{v}_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

Put $\lambda = 90$ in ①

$$-10x + 100y + 40z = 0 \quad \text{--- ①}$$

$$100x + 80y + 140z = 0 \quad \text{--- ②}$$

$$40x + 140y + 110z = 0$$

$$\frac{x}{1} = \frac{-y}{2} = \frac{-z}{3}$$

$$\left| \begin{array}{cc|c} 100 & 40 & 0 \\ 80 & 140 & 0 \end{array} \right| \quad \left| \begin{array}{cc|c} -10 & 40 & 0 \\ 100 & 140 & 0 \end{array} \right| \quad \left| \begin{array}{cc|c} -10 & 100 & 0 \\ 100 & 80 & 0 \end{array} \right|$$

$$= \frac{x}{10800} = \frac{-y}{-5400} = \frac{-z}{-10800}$$

$$= \frac{x}{2} = \frac{y}{1} = \frac{z}{-2}$$

$$v_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \quad \|v_2\| = 3$$

$$\hat{v}_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix}$$

Put $\lambda = 0$ in ①

$$80x + 100y + 40z = 0$$

$$100x + 170y + 140z = 0$$

$$40x + 140y + 200z = 0$$

$$\frac{x}{1} = \frac{-y}{2} = \frac{-z}{3}$$

$$\left| \begin{array}{cc|c} 100 & 40 & 0 \\ 170 & 140 & 0 \end{array} \right| \quad \left| \begin{array}{cc|c} 80 & 40 & 0 \\ 100 & 140 & 0 \end{array} \right| \quad \left| \begin{array}{cc|c} 80 & 100 & 0 \\ 100 & 170 & 0 \end{array} \right|$$

$$\frac{-2}{7200} = \frac{-4}{7200} = \frac{3}{3600}$$

$$= \frac{2}{2} = \frac{-4}{2} = \frac{3}{1}$$

$$v_2 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \quad ||v_3|| = 3 \quad \hat{v}_3 = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}$$

$$V = [\hat{v}_1 \ \hat{v}_2 \ \hat{v}_3] = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$U = [\hat{u}_1 \ \hat{u}_2]$$

$$\hat{u}_1 = \frac{1}{\sigma_1} A \hat{v}_1$$

$$= \frac{1}{6\sqrt{10}} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

$$= \begin{bmatrix} 3/\sqrt{10} \\ 4/\sqrt{10} \end{bmatrix}$$

$$\hat{u}_2 = \frac{1}{\sigma_2} A \hat{v}_2$$

$$= \frac{1}{3\sqrt{10}} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix}$$

$$= \frac{1}{3\sqrt{10}} \begin{bmatrix} -3 \\ 9 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{10} \\ 3/\sqrt{10} \end{bmatrix}$$

$$U = \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 4/\sqrt{10} & 3/\sqrt{10} \end{bmatrix}$$

$$A = U \Sigma V^T$$

$$= \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 4/\sqrt{10} & 3/\sqrt{10} \end{bmatrix} \begin{bmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$$